Mohammed Mahdi Jahangiri

Games Programmer: Gameplay & AI Systems

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Gameplay and AI programmer with shipped VR/XR experience. 2+ years of C#/C++ across Unity and Unreal Engine 5. Shipped a VR title to the Meta Quest Store; led a 3–6 person programming team on a Steam-bound Unity project. Published NPC dialogue research at ICHORA 2024. MSc dissertation focuses on safe AI companions for games.

Technical Skills

Core Languages: C++, C#, Python

Engines & Tools: Unity, Unreal Engine 5, Git, Plastic SCM, Unity DevOps, Creation Kit, Papyrus/PapyrusUtil

AI & Systems: FastAPI, Whisper, Hugging Face, Wit.ai, LLM integration, behaviour trees, NPC dialogue systems

Production: Steam builds, code review, sprint planning, cross-functional coordination, profiling and optimisation

Experience

Programming Team Coordinator — Landell Games
Feb 2025 – Present

Remote / Sweden — Core C# programming team lead on Apoceus, a Unity project shipping to Steam.

- Coordinated a **3–6 person programming team**, translating production priorities into sprint tasks and raising sprint success from **40% to 90%**
- Profiled and **optimised rendering** bottlenecks with the **Unity Profiler**, improving frame time by **10–30 FPS** on target hardware
- Owned weekly **Unity DevOps** branch merges and uploaded builds to a private **Steam** branch
- Mediated **cross-disciplinary** concerns between **5 department leads** (programming, art, design, audio, production)
- Implemented **ScriptableObject**-based unit/data configuration, Unity asset/**LOD** setup and gameplay **state-machine** logic, supporting designer-led balance tweaks while resolving cross-system integration issues

R&D Technical Artist (Contract) — Awkward Studios / Media Cymru
Apr 2026 – Jun 2026

Cardiff — Active R&D for CULTVR Lab: 360° dome-projection game-show installation.

- Solved **360° dome-projection UI distortion** by developing an editor-safe **Unity C#** component that directly manipulated **TextMesh Pro** vertex buffers using trigonometry and **Quaternion** rotations
- Engineered a projection-boundary clipping and reveal system to reduce text artifacting where UI elements crossed half-sphere dome limits
- Built projection-ready material, shader and **particle system** setups for large-scale immersive display constraints
- Programmed modular mini-game state transitions using **C# delegates** and **coroutines**, decoupling sequence logic from UI flow in a **360° game-show format**
- Used **structs** as lightweight data containers for mini-game state and transition data
- Collaborated via **GitHub Pull Request** workflow in a small cross-disciplinary R&D team

Programming Intern — Landell Games
Dec 2024 – Jan 2025

Remote / Sweden

- Delivered gameplay features and bug fixes in **Unity C#** within a remote **Scrum** pipeline using **Git/Plastic** and **Linear**

- Participated in sprint planning and cross-discipline code reviews, supporting clean **Git/Plastic** branch hygiene across a remote team

Selected Projects & Publications

AI-Driven NPC Interaction Prototype
Unreal Engine 5, C++, Python, FastAPI

- Built a local **Python FastAPI** server bridging player microphone input to **Unreal Engine 5** via a custom API
- Integrated **Whisper** (speech-to-text), **Wit.ai** (intent classification) and local **Hugging Face** models for real-time text and voice emotion detection
- Trained the voice-emotion model and configured **Wit.ai** intents using custom voice recordings
- Wired microphone access in **Unreal C++ and Blueprint**, returning predicted intent/emotion signals for real-time NPC **behaviour-tree** logic

VR Playground — Meta Quest Store
Unity, C#, Meta Quest SDK

- Built and tuned the **core gameplay loop** with physics-based standalone headset-only interaction
- Handled the full Meta Quest Store submission, compliance and publishing pipeline
- Iterated post-launch on player feedback, tuning interaction and performance

MSc Thesis — Safe AI Companions for Games
Skyrim, Creation Kit, SKSE

- Built a custom **Skyrim** companion mod using **Creation Kit**, **Papyrus/PapyrusUtil** JSON request/response files and a Python LLM bridge
- Prototyped LLM-driven companion dialogue with bounded context, safety constraints and session/error logging as design priorities
- Evaluating companion AI against branching-dialogue baselines, measuring player interaction quality, latency, safety and reliability

Publication — ICHORA 2024
Balancing Game Satisfaction and Resource Efficiency: LLM and Pursuit Learning Automata for NPC Dialogues

- Published and presented paper proposing a resource-efficient local LLM pipeline for dynamic NPC dialogue
- Applied pursuit learning automata to balance player satisfaction against computational cost

Education

MSc Artificial Intelligence — Cardiff University
2025–2027

Taught component completed June 2026. Dissertation in progress: safe AI companions for games.

- Computer Vision: bounding-box localisation via **RF-DETR** fine-tuning on **VOC2012**
- Computational Linguistics: **NLP** pipeline for voice-driven NPC interaction, including data gathering, labelling, parameter tuning, **speech-to-text**, **intent detection** and **emotion analysis**

BEng Computer Engineering — Islamic Azad University, Tehran
2020–2025

- Graduated with **First-Class Honours**
- Core study: **Data Structures & Algorithms**, **Advanced Programming (C++)**, Computational Architecture and Networks

Awards & Interests

DiffTrailblazers — Cardiff University: Won 1st place at a university-wide innovation competition, receiving a **£1,500** prize for a practical innovation project.

Interests: Judo (green belt) | Mandarin Chinese | Skyrim VR and Modding | XR R&D